

lab-06-solution

Exercise 1

```
# find_distance is a function that finds a distance between points
# on two normal distributions that the user specifies.

find_distance <- function(x, m1, m2, sd1, sd2) {
  0.5 * abs(dnorm(x, mean = m1, sd = sd1) - dnorm(x, mean = m2, sd = sd2))
}
```

Exercise 2

```
ggplot(data = boot_dist, aes(x = boot_props)) +
  geom_histogram(binwidth = 0.01, color = "darkblue", fill = "skyblue") +
  geom_vline(xintercept = c(ex_5_interval$lower, ex_5_interval$upper),
            color = "darkred", lwd = 2) +
  labs(x = "Bootstrap proportions",
       y = "Count",
       title = "Most Durham residents satisfied with parks and recreation",
       subtitle = "1,000 bootstrap replicates with 99% CI shown")
```

Exercise 3

```
# You may have noticed that ncbikecrash isn't a great dataset name.
# Give it a better name in the code chunk below.
# Make sure that everything in the chunk conforms to tidyverse style guidelines,
# including this comment!
```

```
nc_bike_crash |>
  group_by(rural_urban) |>
  count(ambulance_req) |>
  mutate(prob_amb = n / sum(n)) |>
  filter(ambulance_req == "Yes") |>
  select(rural_urban, prob_amb)
```

Exercise 4

```
data <- data.frame(
  group = c("A", "A", "B", "B", "C", "C"),
  value = c(3, 5, 4, 6, 2, 7)
)

ggplot(data, aes(x = group, y = value, fill = group)) +
  geom_bar(stat = "identity") +
  theme_bw() +
  labs(
    title = "Average Value by Group",
    x = "Group",
    y = "Value") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

Exercise 5

```
set.seed(5)
boot_dist <- numeric(1000)
satisfaction <- durham |>
  filter(quality_parks_rec != 9)

for (i in 1:1000) {
  indices <- sample(1:nrow(satisfaction), replace = T)

  boot_prop <- satisfaction |>
    slice(indices) |>
    mutate(parks_satisfied = ifelse(quality_parks_rec > 3, 1, 0)) |>
    summarize(prop_parks = mean(parks_satisfied)) |>
    pull()
}
```

```
    boot_dist[i] <- boot_prop
  }

boot_dist <- tibble(boot_props = boot_dist)

satis_conf_interval <- boot_dist |>
  summarize(lower = quantile(boot_props, 0.005),
            upper = quantile(boot_props, 0.995)) |>
  round(3)
```